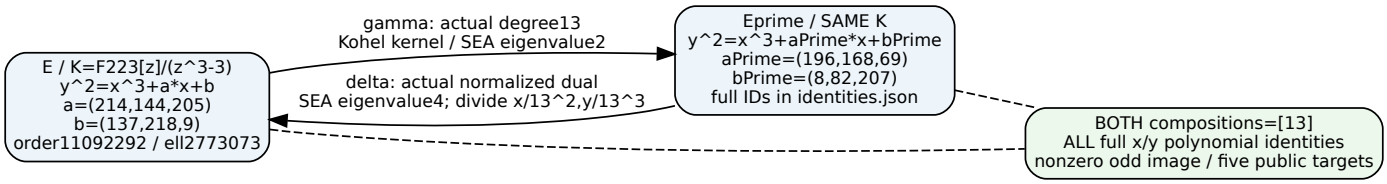
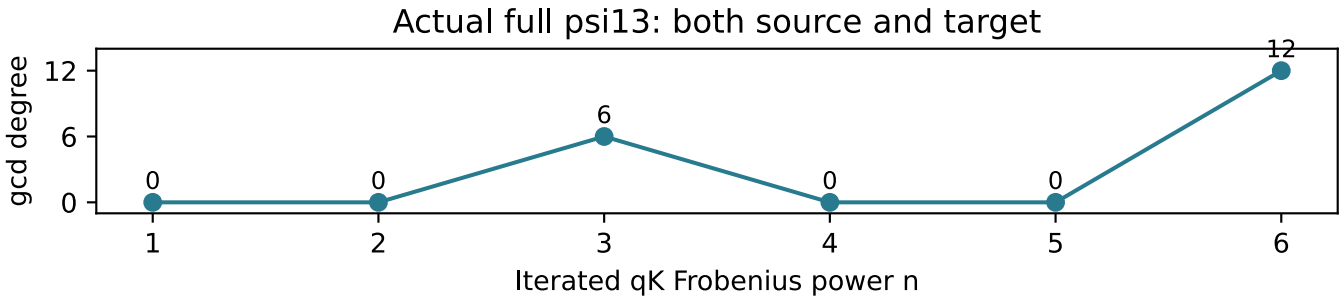
An actual odd source isogeny over the cubic field

2026-11-13 · N4 q223 / seed1 · unsigned / partial formalization





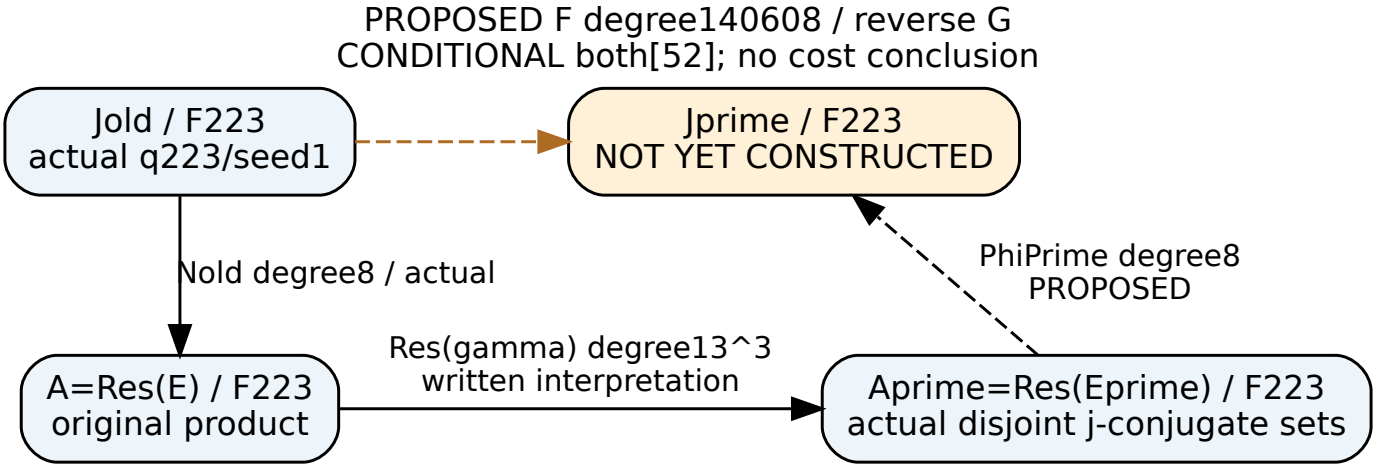
Prospectivee9234b precedes code. Native02/de87bd passes one unfiltered test and all2452 selected hashes remain unchanged. Original field, polynomial, division-polynomial, Kohel, SEA and affine group engines are reused as ordinary dependencies. First build101/1ea2d5 preserved seven nonexistent-square-API errors; the sole repair calls existing original multiplication. No native failure is omitted.

The entire degree84 psi13 is squarefree. Six actual qK Frobenius gcd degrees are0,0,6,0,0,12. D6/D3 gives the degree6 forward kernel/eigenvalue2. On Eprime, D3 gives the normalized dual kernel/eigenvalue4. Their first-power gcds are1, so neither kernel has a K-rational nonidentity point. No unsupported point-extension representation or placeholder generator is used.

For each normalized map xPrime=N/D, yPrime=y*T/D^2, T=Nprime*D-N*Dprime. The FULL identity S*T^2=N^3*D+aPrime*N*D^3+bPrime*D^4 has zero polynomial residual. Each deliberately wrong target b coefficient fails. Numerator/denominator degrees13/12 and gcd1, plus smooth projective extension and O preservation, give actual degree13 homomorphisms in the written argumentb9a91f (Milne AV Cor2.2/Thm3.1).

The normalized dual codomain is E(13^4*a,13^6*b); output scaling x/13^2,y/13^3 gives delta:Eprime→E. BOTH full x AND y composition cross-polynomials agree with multiplication[13], each x degree169/168/coprime. Raw division polynomials retain leading13; y13=y*x13prime/13. Whole coefficients are in all-native-records.json.xz, not replaced by random point evaluations.

A different product; the next Jacobian remains proposed



Native checks retain all36 source and36 target samples, both[13] compositions,1024 homomorphism pairs in each direction, all four rational2-torsion points bijectively, two off-curve rejections, the nonzero exact-ell2773073 generator image and all five unchanged public targets. Known answers are test-only and stay outside runtime map inputs. Source and target equations, field/model tags and full map polynomials are explicit.

Actual source j conjugates are(166,136,127),(166,135,47),(166,175,49); target conjugates(217,167,46), (217,10,10),(217,46,167). They are disjoint. A polarized product isomorphism would have degree-one columns, forcing an elliptic isomorphism between the two sets. The written argument excludes this, so Aprime is a different canonically polarized product. This establishes no computational advantage.

Since both exact rational kernels are trivial, gamma and delta inject finite K-point groups in opposite directions; their orders equal the original11092292 and gamma is bijective. Odd inverse=[639940]delta uses13*639940=1+3*ell. Whole rational inverse=[3413013]delta uses13*3413013=1+4*11092292. These inverse scopes are distinct; no DLP table or scalar fitting is used.

Six faithful exact partial Lean bindings pass both unchanged original4.35/4.34.1 gates/zero trusted: c6dca4 curve-equation clearing,796f92 scaling,7781ea scalar recovery from action/composition/torsion axioms and Bezout,ee0e7e odd13,82763b whole13,caa2b3 proposed52 integer certificates. Ring lemmas use standard propext/Classical.choice/Quot.sound; recovery and integers use no axioms. Whole proof capturesb32320/67342c. Initial caller PATH not-covered78ddb7 is retained; the SAME first claim later passes. Actual curves/morphisms/ degree/dual/finite-group/product interpretations remain written,50 local bindings total.

Next independently construct every HLP outcome for Aprime with fresh typed source/model/origin identities and complete PhiPrime/NormPrime. Only CONDITIONAL on those maps, F=PhiPrime*Res(gamma)*Nold and G=PhiOld*Res(delta)*NormPrime give both[52], degree140608 and odd inverse[159985]G. No new Jacobian or actual[52] map is yet supplied. Original S3 checkpoint/live budget/capacity, complete costs and strongest references remain required for a performance claim.

Prior kernel fullf4c0a7/all617 guards and nine-workflow CI6add5b/all80 native44 canonical digests are settled. New odd-source ci_odd freezes639 portable inputs/640 with manifest, thirteen phases,ALL4 whole JSON,both original6-proof gates/6 bindings; own clean full/CI is pending. New diagrams add the actual odd source and explicitly conditional successor. Previous gluing/kernel diagrams were checked and remain correct.